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Inventor(s)

KIGUCHI; Ryoga et al.

WIRING BOARD AND POWER CONVERSION APPARATUS

Abstract

A wiring board includes a hard substrate provided for each of a plurality of semiconductor elements connected in parallel and a soft substrate configured to connect the hard substrates to each other. All of the hard substrates of each of the plurality of semiconductor elements are connected by the soft substrate.

Inventors: KIGUCHI; Ryoga (Tokyo, JP), TAKUBO; Hiromu (Tokyo, JP)

Applicant: FUJI ELECTRIC CO., LTD. (Kanagawa, JP)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] The present application is a continuation application of U.S. patent application Ser. No. 18/057,950, filed on Nov. 22, 2022, which is based upon and claims priority to Japanese Patent Application No. 2022-013750, filed on Jan. 31, 2022, the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present disclosure relates to a wiring board, etc.

2. Description of the Related Art

[0003] For example, a wiring board for driving multiple semiconductor elements that are connected in parallel, by a single driving circuit, is known (see Patent Document 1).

RELATED-ART DOCUMENTS

Patent Document

[0004] [Patent Document 1] Japanese Laid-Open Patent Application Publication No. 2018-14792

SUMMARY OF THE INVENTION

[0005] According to one aspect of an embodiment, a wiring board includes a pair of hard substrates provided for each of a plurality of semiconductor elements connected in parallel; a soft substrate provided so as to be at least partially sandwiched between all of the pairs of hard substrates; a first electrode configured to connect a control terminal of the semiconductor element and the hard substrate or the soft substrate; a second electrode configured to connect a reference potential terminal of the semiconductor element and the hard substrate or the soft substrate; a first wiring configured to connect in parallel the first electrodes of each of the plurality of semiconductor elements, at least part of the first wiring being provided on the soft substrate; and a second wiring configured to connect in parallel the second electrodes of each of the plurality of semiconductor elements, at least part of the second wiring being provided on the soft substrate.

[0006] According to another aspect of an embodiment, a wiring board includes a hard substrate provided for each of a plurality of semiconductor elements connected in parallel; a soft substrate configured to connect the hard substrates to each other; a first electrode configured to connect a control terminal of the semiconductor element and the hard substrate; a second electrode configured to connect a reference potential terminal of the semiconductor element and the hard substrate; a first wiring configured to connect in parallel the first electrodes of each of the plurality of semiconductor elements, the first wiring being provided on the hard substrate and the soft substrate; and a second wiring configured to connect in parallel the second electrodes of each of the plurality of semiconductor elements, the second wiring being provided on the hard substrate and the soft substrate, wherein all of the hard substrates of each of the plurality of semiconductor elements are connected by the soft substrate.

[0007] According to yet another aspect of an embodiment, a power conversion apparatus includes the above wiring board; and the plurality of semiconductor elements.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a circuit diagram illustrating an example of a power conversion apparatus including a first example of a wiring board;

[0009] FIG. 2 is a perspective view illustrating the first example of a wiring board;

[0010] FIG. **3** is a top view illustrating the first example of a wiring board;
[0011] FIG. **4** is a circuit diagram illustrating one example of a power conversion apparatus including a second example of a wiring board;
[0012] FIG. **5** is a perspective view illustrating a second example of a wiring board;
[0013] FIG. **6** is a top view illustrating the second example of a wiring board;
[0014] FIG. **7** is a cross-sectional view illustrating a third example of a wiring board;
[0015] FIG. **8** is a cross-sectional view illustrating a fourth example of a wiring board;
[0016] FIG. **9** is a cross-sectional view illustrating the fourth example of a wiring board;
[0017] FIG. **10** is a cross-sectional view illustrating the fourth example of a wiring board;
[0018] FIG. **11** is a cross-sectional view illustrating a fifth example of a wiring board;
[0019] FIG. **12** is a cross-sectional view illustrating the fifth example of a wiring board;
[0020] FIG. **13** is a cross-sectional view illustrating the fifth example of a wiring board;
[0021] FIG. **14** is a circuit diagram illustrating one example of a power conversion apparatus including a sixth example of a wiring board;
[0022] FIG. **15** is a circuit diagram illustrating one example of a power conversion apparatus including a seventh example of a wiring board;
[0023] FIG. **16** is a circuit diagram illustrating one example of a power conversion apparatus including an eighth example of a wiring board;
[0024] FIG. **17** is a circuit diagram illustrating one example of a power conversion apparatus including a ninth example of a wiring board;
[0025] FIG. **18** is a circuit diagram of a power conversion apparatus including a tenth example of a wiring board;
[0026] FIG. **19** is a top view illustrating the wiring pattern of a twelfth example of a wiring board;
and
[0027] FIG. **20** is a top view illustrating a wiring pattern according to a comparative example.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0028] In order to prevent a switching deviation of the multiple semiconductor elements that are connected in parallel, it is desirable that the wiring length between the driving circuit and each semiconductor element is configured to be approximately the same.

[0029] However, in Patent Document 1, a substrate for wiring the gate terminal (base terminal) and the source terminal (emitter terminal) is provided for each of the multiple semiconductor elements, and the wiring length between the driving circuit and the gate terminal and the source terminal of each semiconductor element is made approximately the same, according to a structure in which the substrates are connected by connectors and wiring. Therefore, for example, in order to adjust the wiring length, it may be necessary to adopt a route that goes through another board that is different from the board on which the target gate terminal exists, and the wiring length between the driving circuit and the gate terminal of the semiconductor element may become longer. As a result, due to the increase in parasitic inductance associated with the increase in wiring length, it may not be possible to bring out the performance of the semiconductor element, and the increase in parasitic impedance may cause a gate oscillation phenomenon. Further, for example, poor assembly or the addition of mechanical vibration may cause poor contact at the connector fitting. Thus, there is margin for improvement in terms of reliability.

[0030] Therefore, in view of the above problems, it is desirable to provide a technology that can further improve the reliability when driving multiple semiconductor elements that are connected in parallel, by a single driving circuit.

[0031] According to an embodiment of the present disclosure, reliability can be further improved when multiple semiconductor elements that are connected in parallel, are driven by a single driving circuit.

[0032] Embodiments will be described below with reference to the drawings.

First Example of Wiring Board

[0033] A first example of a wiring board **400** according to the present embodiment will be described with reference to FIGS. **1** to **3**.

[0034] FIG. **1** is a circuit diagram illustrating an example of a power conversion apparatus **1** including the first example of the wiring board **400**. FIG. **2** is a perspective view illustrating the first example of the wiring board **400**. FIG. **3** is a top view illustrating the first example of the wiring board **400**.

[0035] Hereafter, explanations may be made by using an orthogonal coordinate system defined by the X, Y and Z axes in the figure. In some cases, the positive X-axis direction and the negative X-axis direction are collectively referred to as the X-axis direction, the positive Y-axis direction and the negative Y-axis direction are collectively referred to as the Y-axis direction, and the positive Z-axis direction and the negative Z-axis direction are collectively referred to as the Z-axis direction.

[0036] The power conversion apparatus **1** converts, for example, the power of a DC power supply PS into a predetermined power (for example, three-phase AC power at a predetermined voltage and a predetermined frequency) and supplies the power to a predetermined load. The power conversion apparatus **1** is mounted on, for example, industrial machinery for factories, moving objects such as railway vehicles, automobiles, and the like.

[0037] The power conversion apparatus **1** has an arm part **10** including a semiconductor switch element SW, and a leg part **20**, which is configured by two arm parts **10** connected in series, provided between the positive and negative lines of the DC power supply PS. For example, the power conversion apparatus **1** is an inverter device capable of outputting three-phase AC power by connecting three leg parts **20** in parallel between the positive and negative lines of the DC power supply PS.

[0038] The arm part **10** includes semiconductor modules **100a** to **100c**, wirings **200a** and **200b**, a driving circuit board **300**, the wiring board **400**, and an electric circuit **500**.

[0039] The semiconductor modules **100a** to **100c** each include the semiconductor switch element SW.

[0040] The semiconductor switch element SW (an example of a semiconductor element) is a voltage-driven type, for example, a MOSFET (Metal Oxide Semiconductor Field Effect Transistor) as illustrated in FIG. **1**. The semiconductor switch element SW may be an IGBT (Insulated Gate Bipolar Transistor) or a HEMT (High Electron Mobility Transistor), etc. The semiconductor switch element SW is configured by, for example, silicon (silicon: Si) as the main material. The semiconductor switch element SW may be configured by a wide bandgap semiconductor material as the main material. The wide bandgap semiconductor materials are, for example, silicon carbide (silicon carbide: SiC), gallium nitride (gallium nitride: GaN), gallium oxide (gallium oxide: Ga.sub.2O.sub.3), carbon (diamond: C), etc.

[0041] For example, as illustrated in FIGS. **2** and **3**, each of the semiconductor modules **100a** to **100c** have an approximately rectangular solid shape with each side along one the X, Y, and Z axes, and are arranged so as to be approximately equally spaced apart in the X direction.

“Approximately” is intended, for example, to allow manufacturing errors, etc., and is used in the same sense below.

[0042] The wirings **200a** and **200b** are used to connect multiple (in this example, three) semiconductor modules **100a** to **100c** (that is, multiple semiconductor switch elements SW) in parallel. Specifically, the wiring **200a** is provided to connect the drain terminals of the multiple semiconductor switch elements SW, that is, the drain electrodes of the multiple semiconductor modules **100a** to **100c**, to each other. The wiring **200b** connects the source terminals of the multiple semiconductor switch elements SW (an example of a reference potential terminal), that is, the source electrodes of the multiple semiconductor modules **100a**, to each other.

[0043] If the semiconductor switch element SW is a bipolar transistor such as an IGBT, the source terminal, the drain terminal, and the gate terminal (an example of a control terminal) are replaced with an emitter terminal (an example of a reference potential terminal), a collector terminal, and a

base terminal (an example of a control terminal), respectively.

[0044] The driving circuit board **300** is a board on which the driving circuits of multiple semiconductor switch elements SW are mounted.

[0045] In the wiring board **400**, wiring for driving the multiple semiconductor modules **100a** to **100c** (that is, multiple semiconductor switch elements SW) by a driving circuit is mounted. The wiring board **400** includes hard substrates **410a** to **410c** (also referred to as a “rigid board”), a soft substrate **420** (also referred to as a “flexible board”), an electrode **450**, electrodes **460a** to **460c**, electrodes **470a** to **470c**, and wirings **480** and **490**.

[0046] The connection and fixation between the wiring board **400** (for example, the hard substrates **410a** to **410c**) and the semiconductor modules **100a** to **100c** may be implemented by any method. For example, the connection and fixation between the wiring board **400** and the semiconductor modules **100a** to **100c** are implemented by means of co-tightening by screws, pressure bonding by applying pressure, etc.

[0047] The hard substrates **410a** to **410c** are provided to correspond to the semiconductor modules **100a** to **100c**, respectively. The hard substrates **410a** to **410c** may be formed of any material. For example, the hard substrates **410a** to **410c** are glass epoxy substrates or the like.

[0048] For example, as illustrated in FIGS. 2 and 3, the hard substrate **410a** is provided on the top surface (that is, the end face in the positive Z-axis direction) of the semiconductor module **100a**. The hard substrate **410a** has a flat plate shape approximately parallel to the X and Y axes and an approximately rectangular shape with each side in one of the X and Y axes directions in a top view. Two hard substrates **410a** are provided. The two hard substrates **410a** function as a pair of hard substrates **410a** as described below.

[0049] For example, as illustrated in FIGS. 2 and 3, the hard substrate **410b** is provided on the top surface (that is, the end face in the positive Z-axis direction) of the semiconductor module **100b**. Similar to the hard substrate **410a**, the hard substrate **410b** has a flat plate shape that is approximately parallel to the X and Y axes, and an approximately rectangular shape with each side in one of the X and Y axes directions in a top view. Similar to the hard substrate **410a**, two hard substrates **410b** are provided. The two hard substrates **410b** function as a pair of hard substrates **410b** as described below.

[0050] For example, as illustrated in FIGS. 2 and 3, the hard substrate **410c** is provided at approximately the center of the top surface (that is, the end face in the positive Z-axis direction) of the semiconductor module **100c** in the X-axis and Y-axis directions. Similar to the hard substrates **410a** and **410b**, the hard substrate **410c** has a flat plate shape that is approximately parallel to the X and Y axes, and has an approximately rectangular shape with each side in one of the X and Y axes directions in a top view. Similar to the hard substrate **410a**, two hard substrates **410b** are provided. The two hard substrates **410b** function as a pair of hard substrates **410c** as described below.

[0051] The soft substrate **420** is provided so as to be sandwiched between all of the pair of hard substrates **410a**, the pair of hard substrates **410b**, and the pair of hard substrates **410c**. The soft substrate **420** can be made of any material. For example, the soft substrate **420** is a polyimide substrate or the like.

[0052] The soft substrate **420**, for example, as illustrated in FIGS. 2 and 3, has an approximately rectangular shape that is approximately parallel to the X and Y axes and whose X axis dimension is longer than the Y axis dimension. Specifically, the soft substrate **420** has a Y-axis dimension approximately equal to or smaller than that of the hard substrates **410a** to **410c** and an X-axis dimension approximately equal to the distance between the end face in the negative X-axis direction of the hard substrate **410a** and the end face of the positive X-axis direction of the hard substrate **410c**. The soft substrate **420** is sandwiched from above and below (that is, the positive Z-axis direction and the negative Z-axis direction) at the end thereof in the negative X-axis direction by the pair of hard substrates **410a**, sandwiched from above and below at the center thereof in the X-axis direction by the pair of hard substrates **410b**, and sandwiched from above and below at the

end thereof in the positive X-axis direction by the pair of hard substrates **410c**. Thus, on the semiconductor module **100a** (the positive Z-axis side), the hard substrate **410a**, the soft substrate **420**, and the hard substrate **410a** are stacked in order from the bottom (negative Z-axis side). Similarly, on the semiconductor module **100b**, the hard substrate **410b**, the soft substrate **420**, and the hard substrate **410b** are stacked in order from the bottom. Similarly, on the semiconductor module **100c**, the hard substrate **410c**, the soft substrate **420**, and the hard substrate **410c** are stacked in order from the bottom. Then, in the X-axis direction, the soft substrate **420** is exposed in the range between the pair of hard substrates **410a** and the pair of hard substrates **410b** that are adjacent to each other, and is also exposed in the range between the pair of hard substrates **410b** and the pair of hard substrates **410c** that are adjacent to each other.

[0053] The soft substrate **420** is electrically connected to at least one of the two hard substrates **410a**. For the connection between the soft substrate **420** and the hard substrate **410a**, for example, a general via hole is used. The via hole may be a through-via, a stacked via, a buried via, etc. The same may be applied to the via hole used for the via connection described below. For the connection between the soft substrate **420** and the hard substrate **410a**, any known method such as copper inlay or copper backfill via may be used.

[0054] The electrode **450** is used to connect the driving circuit board **300** through the electric circuit **500**. For example, as illustrated in FIGS. 2 and 3, the electrode **450** is provided at the center of the hard substrate **410b** in the X-axis direction and at the end of the hard substrate **410b** in the negative Y-axis direction. The electrode **450** includes an electrode **450a** (third electrode) and an electrode **450b**.

[0055] The electrode **460a** (an example of the first electrode) is connected to the gate electrode of the semiconductor module **100a**, that is, the gate terminal of the semiconductor switch element SW included in the semiconductor module **100a**. For example, as illustrated in FIG. 2, the electrode **460a** is provided on the upper hard substrate **410a** where the surface in the positive Z-axis direction of the pair of hard substrates **410a** is exposed. The electrode **460a** may be provided on the hard substrate **410a** on the lower side (negative Z-axis direction) of the pair of hard substrates **410a**, or on the part of the soft substrate **420** between the two hard substrates **410a**.

[0056] For example, as illustrated in FIG. 2 and FIG. 3, the electrode **460a** is provided at a position of the upper hard substrate **410a** toward the negative X-axis direction and the positive Y-axis direction.

[0057] The electrode **460b** (an example of the first electrode) is connected to the gate electrode of the semiconductor module **100b**, that is, the gate terminal of the semiconductor switch element SW included in the semiconductor module **100b**. For example, as illustrated in FIGS. 2 and 3, the electrode **460b** is provided on the upper hard substrate **410b** where the surface in the positive Z-axis direction of the pair of hard substrates **410b** is exposed. Further, the electrode **460b** may be provided on the hard substrate **410b** on the lower side (negative Z-axis direction) of the pair of hard substrates **410b** or on the part of the soft substrate **420** between the two hard substrates **410b**.

[0058] For example, as illustrated in FIG. 2 and FIG. 3, the electrode **460b** is provided at a position of the upper hard substrate **410b** toward the negative X-axis direction and the positive Y-axis direction.

[0059] The electrode **460c** (an example of the first electrode) is connected to the gate electrode of the semiconductor module **100c**, that is, the gate terminal of the semiconductor switch element SW included in the semiconductor module **100c**. For example, as illustrated in FIGS. 2 and 3, the electrode **460c** is provided on the upper hard substrate **410c** where the surface in the positive Z-axis direction of the pair of hard substrates **410c** is exposed. Further, the electrode **460c** may be provided on the hard substrate **410c** on the lower side (negative Z-axis direction) of the pair of hard substrates **410c** or on the part of the soft substrate **420** between the two hard substrates **410c**.

[0060] For example, as illustrated in FIG. 2 and FIG. 3, the electrode **460c** is provided at a position on the upper hard substrate **410c** toward the negative X-axis direction and the positive Y-axis

direction.

[0061] The electrode **470a** (an example of the second electrode) is connected to the source electrode of the semiconductor module **100a**, that is, the source terminal of the semiconductor switch element SW included in the semiconductor module **100a**. For example, as illustrated in FIGS. 2 and 3, the electrode **470a** is provided on the upper hard substrate **410a** where the surface in the positive Z-axis direction of the pair of hard substrates **410a** is exposed. The electrode **470a** may be provided on the hard substrate **410a** on the lower side (negative Z-axis direction) of the pair of hard substrates **410a**, or on the part of the soft substrate **420** between the two hard substrates **410a**.

[0062] For example, as illustrated in FIG. 2 and FIG. 3, the electrode **470a** is provided at a position of the upper hard substrate **410a** toward the negative X-axis direction and the negative Y-axis direction.

[0063] The electrode **470b** (an example of the second electrode) is connected to the source electrode of the semiconductor module **100b**, that is, the source terminal of the semiconductor switch element SW included in the semiconductor module **100b**. For example, as illustrated in FIGS. 2 and 3, the electrode **470b** is provided on the upper hard substrate **410b** where the surface in the positive Z-axis direction of the pair of hard substrates **410b** is exposed. Further, the electrode **470b** may be provided on the hard substrate **410b** on the lower side (negative Z-axis direction) of the pair of hard substrates **410b** or on the part of the soft substrate **420** between the two hard substrates **410b**.

[0064] For example, as illustrated in FIGS. 2 and 3, the electrode **470b** is provided at a position on the upper hard substrate **410b** toward the negative X-axis direction and the negative Y-axis direction.

[0065] The electrode **470c** (an example of the second electrode) is connected to the source electrode of the semiconductor module **100c**, that is, the source terminal of the semiconductor switch element SW included in the semiconductor module **100c**. For example, as illustrated in FIGS. 2 and 3, the electrode **470c** is provided on the upper hard substrate **410c** where the surface in the positive Z-axis direction of the pair of hard substrates **410c** is exposed. Further, the electrode **470c** may be provided on the hard substrate **410c** on the lower side (negative Z-axis direction) of the pair of hard substrates **410c** or on the part of the soft substrate **420** between the two hard substrates **410c**.

[0066] For example, as illustrated in FIG. 2 and FIG. 3, the electrode **470c** is provided at a position of the upper hard substrate **410c** toward the negative X-axis direction and the negative Y-axis direction.

[0067] In addition to the electrodes **460a** to **460c** and the electrodes **470a** to **470c**, the wiring board **400** may be provided with electrodes (auxiliary terminals) for monitoring the state of the respective semiconductor switch elements SW of the semiconductor modules **100a** to **100c**. The auxiliary terminal is, for example, a high-pressure auxiliary terminal used for a proof-pressure test.

[0068] The wiring **480** (an example of the first wiring) connects electrodes **460a** to **460c** in parallel. Specifically, the wiring **480** connects between the electrode **450a** and each of the electrodes **460a** to **460c**. The wiring **480** includes wirings **480a** to **480c**.

[0069] The wiring **480a** is a wiring portion in the wiring **480** connecting between the electrodes **450a** and **460a**. The wiring **480b** is a wiring portion in the wiring **480** connecting between electrodes **450a** and **460b**. The wiring **480c** is a wiring portion in the wiring **480** connecting between the electrodes **450a** and **460c**. The wirings **480a** to **480c** may be provided independently of each other or may be partially common with the electrode **450a** as the starting point.

[0070] The wiring **480** is mainly provided on the soft substrate **420**. The wiring **480** is, for example, primarily arranged on the soft substrate **420** and also provided on at least one of the hard substrates **410a** to **410c** for the purpose of connection with components, electrodes, etc., of the hard substrates **410a** to **410c**. Further, the wiring **480** may be provided only on the soft substrate **420** among the hard substrates **410a** to **410c** and the soft substrate **420**.

[0071] The wiring **490** (an example of the second wiring) connects the electrodes **470a** to **470c** in parallel. Specifically, the wiring **490** connects between the electrode **450b** and each of the electrodes **470a** to **470c**. The wiring **490** includes wirings **490a** to **490c**.

[0072] The wiring **490a** is a wiring portion in the wiring **490** connecting between the electrodes **450b** and **470a**. The wiring **490b** is a wiring portion in the wiring **490** connecting between the electrodes **450b** and **470b**. The wiring **490c** is a wiring portion in the wiring **490** connecting between the electrodes **450b** and **470c**. The wirings **490a** to **490c** may be provided independently of each other or may be partially common with the electrode **450b** as the starting point.

[0073] The wiring **490** is provided, for example, on the soft substrate **420** and at least one of the hard substrates **410a** to **410c**. Further, the wiring **490** may be provided only on the soft substrate **420** among the hard substrates **410a** to **410c** and the soft substrate **420**.

[0074] The electric circuit **500** connects between the driving circuit board **300** and the electrode **450** of the wiring board **400**. The electrical circuit **500** includes electrical circuits **500a** and **500b**.

[0075] The electric circuit **500a** connects between the driving circuit board **300** and the electrode **450a** of the wiring board **400**. The electric circuit **500b** connects between the driving circuit board **300** and the electrode **450b** of the wiring board **400**.

[0076] Thus, the wiring board **400** is configured as a rigid flexible board with a combination of the hard substrates **410a** to **410c** and the soft substrate **420**. Specifically, part of the wiring board **400** has a stacked structure in which each of the pair of hard substrates **410a**, the pair of hard substrates **410b**, and the pair of hard substrates **410c** is an outer layer and the soft substrate **420** is an inner layer. The hard substrates **410a** to **410c** for each of the multiple semiconductor modules **100a** to **100c** can be connected to each other in such a way that the soft substrate **420** is sandwiched between the two hard substrates **410a**, between the two hard substrates **410b**, and between the two hard substrates **410c**.

[0077] For example, when hard substrates **410a** to **410b** are electrically connected to each other by a connection method using connectors, the wiring lengths between the electrode **450a** and each of the electrodes **460a** to **460c** are matched, and, therefore, the absolute value of the wiring length may become large. This is because it is necessary to adjust the length of the path by running the wiring outward and back along the connecting path by the connector. As a result, the performance of the semiconductor switch element SW may not be fully brought out due to the increase in parasitic inductance caused by the increase in the wiring length. Further, a gate oscillation phenomenon may occur due to an increase in parasitic impedance due to an increase in the wiring length. Further, poor connection may occur in the connector fitting due to poor assembly or mechanical vibration. In particular, when the power conversion apparatus **1** is mounted on a moving object such as a railway vehicle or an automobile, mechanical vibration is likely to be applied, which may increase the concern of poor connection. Connectors can also lead to an increase in the dimension in the height direction (Z-axis direction), which may hinder an attempt of reducing the dimension in the height direction (lowering the height). In particular, in the case of automobiles, the space for mounting components is extremely limited and there is a high need for a low height, and, therefore, there may be a significant impact on the ability to be mounted on automobiles.

[0078] In contrast, in this example, the wiring length between the electrode **450a** and each of the electrodes **460a** to **460c** corresponding to the gate electrode for each of the semiconductor modules **100a** to **100c** can be adjusted by the wiring pattern in the hard substrates **410a** to **410c** and the soft substrate **420**. Therefore, the degree of freedom of the wiring pattern makes it possible to prevent an increase in the wiring length when matching the wiring length between the electrode **450a** and each of the electrodes **460a** to **460c**. As a result, the performance of the semiconductor switch element SW can be brought out to a desired level and the occurrence of gate oscillation phenomenon can be prevented. Moreover, connectors are not used for the interconnection of the hard substrates **410a** to **410c**, and, therefore, the occurrence of connection failure problems can be

prevented and the height of the wiring board **400** can be reduced. Therefore, the wiring board **400** with higher reliability and operation stability can be implemented. Moreover, a manufacturing process for fitting a connector is not required, and labor saving in assembly work, improvement in manufacturing efficiency, reduction in manufacturing cost and the like can be achieved.

Second Example of Wiring Board

[0079] Next, a second example of the wiring board **400** according to the present embodiment will be described with reference to FIGS. **4** to **6**.

[0080] Hereafter, the explanation will focus on the parts different from the first example above, and the same or corresponding content as the first example above may be omitted.

[0081] FIG. **4** is a circuit diagram illustrating an example of the power conversion apparatus **1** including a second example of the wiring board **400**. FIG. **5** is a perspective view illustrating a second example of the wiring board **400**. FIG. **6** is a top view illustrating a second example of the wiring board **400**.

[0082] The wiring board **400** includes hard substrates **430a** to **430c**, soft substrates **440a** and **440b**, the electrode **450**, the electrodes **460a** to **460c**, the electrodes **470a** to **470c**, and the wirings **480** and **490**.

[0083] The hard substrates **430a** to **430c** are provided to correspond to the semiconductor modules **100a** to **100c**, respectively.

[0084] For example, as illustrated in FIGS. **5** and **6**, the hard substrate **430a** is provided on the top surface (that is, the positive Z-axis end face) of the semiconductor module **100a**. The hard substrate **430a** has a flat plate shape approximately parallel to the X and Y axes and an approximately rectangular shape with each side in one of the X and Y axes directions in a top view.

[0085] For example, as illustrated in FIGS. **5** and **6**, the hard substrate **430b** is provided on the top surface (that is, the positive Z-axis end face) of the semiconductor module **100b**. The hard substrate **430b**, similar to the hard substrate **430a**, has a flat plate shape that is approximately parallel to the X and Y axes, and an approximately rectangular shape with each side in one of the X and Y axes directions in a top view.

[0086] For example, as illustrated in FIGS. **5** and **6**, the hard substrate **430c** is provided at approximately the center of the top surface (that is, the positive Z-axis end face) of the semiconductor module **100c** in the X-axis and Y-axis directions. The hard substrate **430c**, similar to the hard substrates **430a** and **430b**, has a flat plate shape that is approximately parallel to the X and Y axes, and has an approximately rectangular shape with each side in one of the X and Y axes directions in a top view.

[0087] Each of the ends of the soft substrate **440a** is electrically connected to one of the hard substrates **430a** and **430b** so as to connect between the hard substrates **430a** and **430b**. For example, as illustrated in FIG. **5** and FIG. **6**, the soft substrate **440a** has a flat plate shape approximately parallel to the X and Y axes and has an approximately rectangular shape with each side extending in one of the X and Y axes directions in a top view. In the soft substrate **440a**, the negative X-axis end is electrically connected to the upper surface (the positive Z-axis end) of the hard substrate **430a**, and the positive X-axis end is electrically connected to the upper surface of the hard substrate **430b**.

[0088] Each of the ends of the soft substrate **440b** is electrically connected to one of the hard substrates **430b** and **430c** so as to connect between the hard substrates **430b** and **430c**. For example, as illustrated in FIGS. **5** and **6**, the soft substrate **440b** has a flat plate shape approximately parallel to the X and Y axes and an approximately rectangular shape with each side extending in the X and Y axes directions in a top view. In the soft substrate **440b**, the negative X-axis end is electrically connected to the upper surface of the hard substrate **430b** (the positive Z-axis end surface), and the positive X-axis end is electrically connected to the upper surface of the hard substrate **430c**.

[0089] The electrical connection between the soft substrate **440a** and the hard substrates **430a** and **430b** may be implemented by any joining method. The electrical connection between the soft

substrate **440a** and the hard substrates **430a** and **430b** is implemented, for example, by joining with solder or the like. The same may be applied to the electrical connection between the soft substrate **440b** and the hard substrates **430b** and **430c**. Further, mechanical connection (holding) between the soft substrate **440a** and the hard substrates **430a** and **430b** may be implemented by any method. Mechanical connection (holding) between the soft substrate **440a** and the hard substrates **430a** and **430b** is implemented, for example, by joining with solder, resin, etc. The same may be applied to the mechanical connection (holding) between the soft substrate **440b** and the hard substrates **430b** and **430c**.

[0090] The wiring **480** is arranged on the hard substrates **430a** to **430c** and the soft substrates **440a** and **440b**. The same applies to the wiring **490**.

[0091] Thus, in this example, the wiring board **400** has a configuration equivalent to that of a rigid flexible board formed by a combination of the hard substrates **430a** to **430c** and the soft substrates **440a** and **440b**. Specifically, the wiring board **400** has a structure in which the hard substrates **430a** to **430c** corresponding to the multiple semiconductor modules **100a** to **100c**, respectively, are electrically connected to each other by the soft substrates **440a** and **440b**.

[0092] Thus, in this example, the wiring length between the electrode **450a** and each of the electrodes **460a** to **460c** can be adjusted according to the wiring patterns in the hard substrates **430a** to **430c** and the soft substrates **440a** and **440b**. Therefore, as in the case of the first example above, the degree of freedom of the wiring pattern makes it possible to prevent an increase in the wiring length when matching the wiring length between the electrode **450a** and each of the electrodes **460a** to **460c**. As a result, the performance of the semiconductor switch element SW can be brought out to a desired level and the occurrence of gate oscillation phenomenon can be prevented.

Moreover, connectors are not used for the interconnection of the hard substrates **410a** to **410c**, and, therefore, the occurrence of connection failure problems can be prevented and the height of the wiring board **400** can be reduced. Therefore, the wiring board **400** with higher reliability and operation stability can be implemented. Moreover, a manufacturing process for fitting a connector is not required, and labor saving in assembly work, improvement in manufacturing efficiency, reduction in manufacturing cost and the like can be achieved. Moreover, the hard substrates **430a** to **430c** and the soft substrates **440a** and **440b** need only be joined by known methods, and, therefore, the adoption of simpler processes can further promote labor-saving of assembly work, improvement of manufacturing efficiency, and reduction of manufacturing cost.

Third Example of Wiring Board

[0093] Next, a third example of the wiring board **400** according to the present embodiment will be described with reference to FIG. 7.

[0094] Hereafter, the explanation will focus on the parts that are different from examples 1 and 2 above, and the same or corresponding content as examples 1 and 2 above may be omitted. In this example, the circuit diagram of the power conversion apparatus **1** including the wiring board **400**, the perspective view of the wiring board **400**, and the top view of the wiring board **400** are the same as in the above second example, so the illustration is omitted, and the explanation is made by referring to FIGS. 4 to 6 of the above second example.

[0095] FIG. 7 is a cross-sectional view illustrating a third example of the wiring board **400**. Specifically, FIG. 7 is a cross section of the line F-F' in FIG. 5.

[0096] In this example, the hard substrates **430a** to **430c** have the same structure as each other, and, therefore, a description of the hard substrate **430c** will be mainly given as a representative example, and illustrations and descriptions of the hard substrates **430a** and **430b** will be omitted.

[0097] As illustrated in FIG. 7, the hard substrate **430c** is a multilayer substrate including insulating layers **431A** to **431C** and a protective film **435**.

[0098] The insulating layers **431A** to **431C** are stacked in the stated order from the top (that is, from the positive Z-axis direction). Further, wiring layers are provided on the surface of the insulating layer **431A** in the positive Z-axis direction, between the insulating layers **431A** and **431B**

(the surface of the insulating layer **431B** in the positive Z-axis direction), between the insulating layers **431B** and **431C** (the surface of the insulating layer **431B** in the negative Z-axis direction), and on the surface of the insulating layer **431C** in the negative Z-axis direction. That is, four stacked wiring layers are provided in the hard substrate **430c**.

[0099] The protective film **435** is provided on the surface of the insulating layer **431A** in the positive Z-axis direction and the surface of the insulating layer **431C** in the negative Z-axis direction. The protective film **435** is, for example, a solder resist.

[0100] As illustrated in FIG. 7, in this example, the wiring **480** includes wirings **480A** and **480B** as portions provided in the hard substrate **430c**. In this example, the wiring **490** includes wirings **490A** to **490C** as portions provided in the hard substrate **430c**.

[0101] The wiring **480A** is provided, among the inner layers of the hard substrate **430c**, on the inner layer in the positive Z-axis direction (hereafter, for the sake of convenience, the “first inner layer”), that is, on the surface of the insulating layer **431B** in the positive Z-axis direction between the insulating layers **431A** and **431B**. For example, as illustrated in FIG. 7, the wiring **480A** is arranged at the center in the Y-axis direction.

[0102] The wiring **480B** is provided, among the inner layers of the hard substrate **430c**, on the inner layer in the negative Z-axis side (hereafter, for the sake of convenience, the “second inner layer”), that is, on the surface of the insulating layer **431B** in the negative Z-axis direction between the insulating layers **431B** and **431C**. For example, as illustrated in FIG. 7, similar to the wiring **480A**, the wiring **480B** is located at the center in the Y-axis direction.

[0103] The electrical connection between the wirings **480A** and **480B** is implemented by a connection part **481**. The connection part **481** may implement an electrical connection between the wirings **480A** and **480B** by any method. The connection part **481** implements an electrical connection between the wirings **480A** and **480B** by, for example, via connection. Further, the connection part **481** may implement an electrical connection between the wirings **480A** and **480B** by copper inlay or the like.

[0104] The wiring **490A** is provided on the outer layer of the hard substrate **430c** in the positive Z-axis direction, that is, on the surface of the insulating layer **431A** in the positive Z-axis direction. The wiring **490A** is provided in a range including the entire area of the wirings **480A** and **480B** in the Y-axis direction.

[0105] The wiring **490B** is provided in the first inner layer of the hard substrate **430c**. The wiring **490B** is located both between the wiring **480A** and the positive Y-axis end of the hard substrate **430c** and between the wiring **480A** and the negative Y-axis end of the hard substrate **430c**. That is, the wiring **490B** is arranged to sandwich the wiring **480A** in the Y-axis direction. For example, as illustrated in FIG. 7, the wiring **490B** is arranged so that the position of the positive Y-axis end in the Y-axis direction approximately coincides with the position of the positive Y-axis end of the wiring **490A** and the wiring **490D** in the Y-axis direction. The same may apply to the position at the negative Y-axis end.

[0106] The wiring **490C** is provided in the second inner layer of the hard substrate **430c**. The wiring **490C** is arranged both between the wiring **480B** and the positive Y-axis end and between wiring **480B** and the negative Y-axis end. That is, the wiring **490C** is arranged to sandwich the wiring **480B** in the Y-axis direction. For example, as illustrated in FIG. 7, the wiring **490C** is arranged so that the position of the positive Y-axis end in the Y-axis direction approximately coincides with the position of the positive Y-axis end of the wiring **490A** and the wiring **490D** in the Y-axis direction. The same may apply to the position at the negative Y-axis end.

[0107] The wiring **490D** is provided on the outer layer of the hard substrate **430c** in the negative Z-axis direction, that is, on the surface of the insulating layer **431C** in the negative Z-axis direction. The wiring **490D** is provided in a range including the entire area of the wirings **480A** and **480B** in the Y-axis direction. For example, as illustrated in FIG. 7, the wiring **490D** is arranged so that the position of the positive Y-axis end approximately coincides with the position of the positive Y-axis

end of the wiring **490A**. The same may apply to the position at the negative Y-axis end.

[0108] The electrical connection of the wirings **490A** to **490D** is implemented by a connection part **491**. The connection part **491** may implement electrical connection of the wirings **490A** to **480D** by any method. The connection part **481** implements an electrical connection between the wirings **490A** to **490D** by, for example, via connection. Further, the connection part **491** may implement an electrical connection between the wirings **490A** to **490D** by copper inlay or the like.

[0109] For example, as illustrated in FIG. 7, the connection part **491** is provided on both the negative Y-axis side and the positive Y-axis side of the wirings **480A** and **480B** and the connection part **481** in the Y-axis direction.

[0110] Thus, in this example, the wiring **480** is arranged to be sandwiched between the wirings **490B** and **490C** in the Y-axis direction and between the wirings **490A** and **490D** in the Z-axis direction. The connection part **481** for connecting the layers of the wiring **480** may be arranged to be sandwiched between the connection parts **491** for connecting the layers of the wiring **490** in the Y-axis direction. Thus, the wiring **480** can be arranged so as to be surrounded by the wiring **490** when viewed from the X-axis direction in the hard substrate **430c**. Therefore, the wiring **480** is electromagnetically shielded by the wiring **490** and the connection part **491**, and as a result, the parasitic inductance component of the wiring **490** (the wirings **490A** to **490D**) of the hard substrate **430c** can be prevented (canceled). Therefore, the gate oscillation phenomenon caused by the parasitic component of the wirings **480** and **490** and the imbalance of the current flowing through the semiconductor switch element SW can be prevented, and the wiring board **400** with higher reliability and operation stability can be implemented.

Fourth Example of Wiring Board

[0111] Next, a fourth example of the wiring board **400** according to the present embodiment will be described with reference to FIGS. 8 to 10.

[0112] The following discussion is focused on the parts that differ from the first to third examples above, and the same or corresponding content as the first to third examples above may be omitted. Further, in this example, the circuit diagram of the power conversion apparatus **1** including the wiring board **400**, the perspective view of the wiring board **400**, and the top view of the wiring board **400** are the same as in the first example described above, so the illustration is omitted, and the explanation is made by referring to FIGS. 1 to 3 of the first example described above.

[0113] FIGS. 8 to 10 are cross-sectional views illustrating the fourth example of the wiring board **400**. Specifically, FIGS. 8 to 10 are cross-sectional views of lines A-A', B-B', and C-C' in FIG. 2.

[0114] A cross-sectional view in a plane parallel to the Y-axis and Z-axis of the soft substrate **420** at a point between the pair of hard substrates **410a** and the pair of hard substrates **410b** in the X-axis direction would be illustrated in the same manner as in FIG. 9, and therefore, the illustration is omitted. Further, a cross-sectional view in a plane parallel to the Y-axis and Z-axis of the wiring board **400** at a point where the soft substrate **420** is sandwiched by the two hard substrates **410a** would be illustrated in the same manner as in FIG. 10, and therefore, the illustration is omitted. Specifically, this would be illustrated by a diagram in which a pair of hard substrates **410c**, a connection part **482c**, and a connection part **492c** in FIG. 10 are each replaced by a pair of hard substrates **410a**, a connection part **482a**, and a connection part **492a**. Further, a cross-sectional view in a plane parallel to the Y and Z axes of the wiring board **400** at a point where the soft substrate **420** is sandwiched by two hard substrates **410b** would be illustrated in the same manner as in FIG. 10, and therefore, the illustration is omitted. Specifically, this would be illustrated by a diagram in which the pair of hard substrates **410c**, the connection part **482c**, and the connection part **492c** in FIG. 10 are each replaced by a pair of hard substrates **410b**, a connection part **482b**, and a connection part **492b**.

[0115] As illustrated in FIGS. 8 to 10, the soft substrate **420** includes insulating layers **421A** to **421E**.

[0116] The insulating layers **421A** to **421E** are stacked in the stated order from the top (that is, from

the positive Z-axis direction). Further, four stacked wiring layers are provided between the insulating layers **421A** and **421B** (the surface of the insulating layer **421B** in the positive Z-axis direction), between the insulating layers **421B** and **421C** (the surface of the insulating layer **421C** in the positive Z-axis direction), between the insulating layers **421C** and **421D** (the surface of the insulating layer **421C** in the negative Z-axis direction), and between the insulating layers **421D** and **421E** (that is, the surface of the insulating layer **421D** in the negative Z-axis direction).

[0117] In this example, the wiring **480** includes wirings **480C** and **480D** as portions provided in the soft substrate **420**.

[0118] The wiring **480C** is provided, among the inner layers of the soft substrate **420**, on the inner layer in the positive Z-axis direction (hereafter, for the sake of convenience, the “first inner layer”), that is, on the surface of the insulating layer **421C** in the positive Z-axis direction between the insulating layers **421B** and **421C**. For example, as illustrated in FIG. **8**, the wiring **480C** is arranged approximately at the center in the X-axis direction in the first inner layer of the soft substrate **420**. Specifically, the wiring **480C** may be arranged in the first inner layer such that the positive X-axis end extends to the position in the X-axis direction where the hard substrate **410c** exists and the negative X-axis end extends to the position in the X-axis direction where the hard substrate **410a** exists. Further, for example, as illustrated in FIG. **9** and FIG. **10**, the wiring **480C** is arranged at the center in the Y-axis direction in the first inner layer.

[0119] The wiring **480D** is provided, among the inner layers of the soft substrate **420**, on the inner layer in the negative Z-axis direction (hereafter, for the sake of convenience, the “second inner layer”), that is, on the surface of the insulating layer **421C** in the negative Z-axis direction between the insulating layers **421C** and **421D**. For example, as illustrated in FIG. **8**, the wiring **480D** is located approximately at the center in the X-axis direction in the second inner layer of the soft substrate **420**. Specifically, the wiring **480D**, similar to the wiring **480C**, may be arranged in the second inner layer such that the positive X-axis end extends to the position in the X-axis direction where the hard substrate **410c** exists and the negative X-axis end extends to the position in the X-axis direction where the hard substrate **410a** exists. For example, as illustrated in FIG. **9** and FIG. **10**, the wiring **480D**, similar to the wiring **480C**, is located at the center of the second inner layer in the Y-axis direction.

[0120] The electrical connection between the wirings **480C** and **480D** and the pair of hard substrates **410a** is implemented by the connection part **482a**. For example, as illustrated in FIG. **8**, the connection part **482a** connects wirings **480C** and **480D** and the pair of hard substrates **410a** in the Z-axis direction at the negative X-axis end of the wirings **480C** and **480D**.

[0121] The electrical connection between the wirings **480C** and **480D** and the pair of hard substrates **410b** is implemented by the connection part **482b**. For example, as illustrated in FIG. **8**, two connection parts **482b** are provided at the center of the pair of hard substrates **410b** in the X-axis direction, and each connects the wirings **480C** and **480D** and the pair of hard substrates **410b** in the Z-axis direction.

[0122] The electrical connection between wirings **480C** and **480D** and the pair of hard substrates **410c** is implemented by the connection part **482c**. For example, as illustrated in FIG. **8**, the connection part **482c** connects the wirings **480C** and **480D** and the pair of hard substrates **410c** in the Z-axis direction at the positive X-axis end of the wirings **480C** and **480D**.

[0123] The connection part **482a** may implement electrical connection between the wirings **480C** and **480D** and the pair of hard substrates **410a** by any method. The connection part **482a** implements electrical connection between the wirings **480C** and **480D** and the pair of hard substrates **410a** by, for example, via connection. Further, the connection part **482a** may implement electrical connection between the wirings **480C** and **480D** and the pair of hard substrates **410a** by copper inlay or the like. The same may be applied to the connection parts **482b** and **482c**.

[0124] The wiring **490** includes wirings **490E** to **490H** as portions provided in the soft substrate **420**.

[0125] The wiring **490E** is provided, among the outer layers of the soft substrate **420**, on the outer layer of in the positive Z-axis direction, that is, on the surface of the insulating layer **421B** in the positive Z-axis direction between the insulating layers **421A** and **421B**. The wiring **490E** is provided in both the X-axis direction and the Y-axis direction of the outer layer in the positive Z-axis direction of the soft substrate **420** in a range including the entire area of the wirings **480C** and **480D**. For example, as illustrated in FIG. **8**, the wiring **490E** is provided in a range from the positive X-axis end to the negative X-axis end in the outer layer in the positive Z-axis direction of the soft substrate **420**.

[0126] The wiring **490F** is provided in the first inner layer of the soft substrate **420**. The wiring **490F** is arranged in the first inner layer of the soft substrate **420**, both between the wiring **480C** and the positive X-axis end of the soft substrate **420** and between the wiring **480C** and the negative X-axis end of the soft substrate **420**. That is, the wiring **490F** is arranged to sandwich the wiring **480C** in the X-axis direction of the first inner layer of the soft substrate **420**. For example, as illustrated in FIG. **8**, the wiring **490F** is arranged at the positive X-axis end and the negative X-axis end in the first inner layer of the soft substrate **420**. Further, the wiring **490F** is arranged in the first inner layer of the soft substrate **420** both between the wiring **480C** and the positive Y-axis end of the soft substrate **420** and between the wiring **480C** and the negative Y-axis end of the soft substrate **420**. That is, the wiring **490F** is arranged to sandwich the wiring **480C** in the Y-axis direction of the first inner layer of the soft substrate **420**. For example, as illustrated in FIG. **9** and FIG. **10**, the wiring **490F** is arranged so that the position of the positive Y-axis end in the Y-axis direction approximately coincides with the position of the positive Y-axis end of the wiring **490E** and the wiring **490H** in the Y-axis direction. The same may apply to the position of the end in the negative Y-axis direction.

[0127] The wiring **490G** is provided in the second inner layer of the soft substrate **420**. The wiring **490G** is arranged in the second inner layer of the soft substrate **420** both between wiring **480D** and the positive X-axis end of the soft substrate **420** and between wiring **480D** and the negative X-axis end of the soft substrate **420**. That is, the wiring **490G** is arranged to sandwich the wiring **480D** in the X-axis direction of the second inner layer of the soft substrate **420**. For example, as illustrated in FIG. **8**, the wiring **490G** is arranged at the positive X-axis end and the negative X-axis end in the first inner layer of the soft substrate **420**, similar to the wiring **490F**. Further, the wiring **490G** is arranged both between the wiring **480D** and the end in the positive Y-axis direction and between the wiring **480D** and the end in the negative Y-axis direction in the second inner layer of the soft substrate **420**. That is, the wiring **490G** is arranged to sandwich the wiring **480D** in the Y-axis direction of the second inner layer of the soft substrate **420**. For example, as illustrated in FIG. **9** and FIG. **10**, the wiring **490G**, similar to the wiring **490F**, is arranged so that the position of the positive Y-axis end in the Y-axis direction approximately coincides with the position of the positive Y-axis end of the wiring **490E** and the wiring **490H** in the Y-axis direction. The same may be applied to the position of the end in the negative direction of the Y-axis.

[0128] The wiring **490H** is provided on the outer layer of the soft substrate **420** in the negative Z-axis direction, that is, on the surface of the insulating layer **421D** in the negative Z-axis direction. The wiring **490H**, similar to the wiring **490E**, is provided in a range including the entire area of the wirings **480C** and **480D** in both the X-axis direction and the Y-axis direction of the outer layer of the soft substrate **420** in the negative Z-axis direction. For example, as illustrated in FIG. **8**, the wiring **490H** is provided in the outer layer in the negative Z-axis direction of the soft substrate **420** in a range from the positive X-axis direction end to the negative X-axis direction end. For example, as illustrated in FIG. **9** and FIG. **10**, the wiring **490H** is arranged so that the end of the wiring **490H** in the positive Y-axis direction approximately coincides with the position of the end of the wiring **490E** in the positive Y-axis direction. The same may also apply to the position in the negative Y-axis direction.

[0129] The electrical connection between the wirings **490E** to **490H** and the pair of hard substrates

410a is implemented by a connection part **492a**. For example, as illustrated in FIG. 10, the connection part **492a** connects the wirings **490E** to **490H** and the pair of hard substrates **410a** in the Z-axis direction at the negative X-axis end of wirings **490E** and **490H**. For example, as illustrated in FIG. 10, the connection part **492a** connects the wirings **490E** to **490H** and the pair of hard substrates **410a** in the Z-axis direction at both ends of the wirings **490E** and **490H** in the positive Y-axis direction and the negative Y-axis direction.

[0130] The electrical connection between the wirings **490E** and **490H** and the pair of hard substrates **410b** is implemented by a connection part **492b**. For example, as illustrated in FIG. 8, the connection parts **492b** are respectively provided at the two ends in the X-axis direction of the pair of hard substrates **410b**, and connect the wirings **490E** to **490H** and the pair of hard substrates **410b** in the Z-axis direction. For example, as illustrated in FIG. 10, the connection part **492b** connects the wirings **490E** to **490H** and the pair of hard substrates **410b** in the Z-axis direction at both ends of the wirings **490E** and **490H** in the positive Y-axis direction and the negative Y-axis direction.

[0131] The electrical connection between the wirings **490E** to **490H** and the pair of hard substrates **410c** is implemented by a connection part **492c**. For example, as illustrated in FIG. 8, the connection part **492c** connects the wirings **490E** to **490H** and the pair of hard substrates **410c** in the Z-axis direction at the positive X-axis end of the wirings **490E** and **490H**. For example, as illustrated in FIG. 10, the connection part **492c** connects the wirings **490E** to **490H** and the pair of hard substrates **410c** in the Z-axis direction at both ends of the wirings **490E** and **490H** in the positive Y-axis direction and the negative Y-axis direction.

[0132] The connection part **492a** may implement electrical connection between the wirings **490E** to **490H** and the pair of hard substrates **410a** by any method. The connection part **492a** implements electrical connection between the wirings **490E** to **490D** and the pair of hard substrates **410a** by, for example, via connection. Further, the connection part **492a** may implement the electrical connection between the wirings **490E** to **490H** and the pair of hard substrates **410a** by copper inlay or the like. The same may be applied to the connection parts **492b** and **492c**.

[0133] Thus, in this example, the wiring **480** is arranged to be sandwiched between the wirings **490F** and **490G** in the X-axis direction and the Y-axis direction, and between the wirings **490E** and **490H** in the Z-axis direction. Thus, the wiring **480** can be arranged so as to be surrounded by the wiring **490** in the soft substrate **420**. Therefore, the wiring **480** is electromagnetically shielded by the wiring **490**, and as a result, the parasitic inductance component of the wiring **490** (the wirings **490E** to **490H**) of the soft substrate **420** can be prevented (canceled). Therefore, the gate oscillation phenomenon caused by the parasitic component of the wirings **480** and **490** and the imbalance of the current flowing through the semiconductor switch element SW can be prevented, and the wiring board **400** with higher reliability and operation stability can be implemented.

Fifth Example of Wiring Board

[0134] Next, the wiring board **400** according to the present embodiment will be described with reference to FIGS. 11 to 13.

[0135] The following discussion is focused on the different parts from the above first to fourth examples, and the same or corresponding content as the above first to fourth examples be omitted. Further, in this example, the circuit diagram of the power conversion apparatus **1** including the wiring board **400**, the perspective view of the wiring board **400**, and the top view of the wiring board **400** are the same as those in the first example described above, so the illustrations are omitted, and the explanation is made by referring to FIGS. 1 to 3 of the first example described above.

[0136] FIGS. 11 to 13 are cross-sectional views illustrating the fourth example of the wiring board **400**. Specifically, FIGS. 11 to 13 are cross-sectional views of lines A-A', B-B' and C-C' in FIG. 2.

[0137] A cross-sectional view in a plane parallel to the Y-axis and Z-axis of the soft substrate **420** at a point between the pair of hard substrates **410a** and the pair of hard substrates **410b** in the X-axis

direction would be illustrated in the same manner as in FIG. 12, and therefore, the illustration is omitted. Further, a cross-sectional view in a plane parallel to the Y-axis and Z-axis of the wiring board **400** at a point where the soft substrate **420** is sandwiched by the two hard substrates **410a** would be illustrated in the same manner as in FIG. 13, and therefore, the illustration is omitted. Specifically, this would be illustrated by a diagram in which a pair of hard substrates **410c**, a connection part **483c**, and a connection part **493c** in FIG. 13 are each replaced by a pair of hard substrates **410a**, a connection part **483a**, and a connection part **493a**. Further, a cross-sectional view in a plane parallel to the Y and Z axes of the wiring board **400** at a point where the soft substrate **420** is sandwiched by two hard substrates **410b** would be illustrated in the same manner as in FIG. 13, and therefore, the illustration is omitted. Specifically, this would be illustrated by a diagram in which the pair of hard substrates **410c**, the connection part **483c**, and the connection part **493c** in FIG. 13 are each replaced by a pair of hard substrates **410b**, a connection part **483b**, and a connection part **493b**.

[0138] As illustrated in FIG. 11 to FIG. 13, the soft substrate **420** includes the insulating layers **421A** to **421E**, as in the case of the fourth example above, and four stacked wiring layers are provided.

[0139] In this example, the wiring **480** includes wirings **480E** and **480F** as a part provided on the soft substrate **420**.

[0140] The wiring **480E** is provided on the first inner layer of the soft substrate **420**, i.e., on the surface of the insulating layer **421C** between the insulating layers **421B** and **421C**. For example, as illustrated in FIG. 11, the wiring **480E** is provided in a range from the positive X-axis end to the negative X-axis end of the soft substrate **420**, in the first inner layer of the soft substrate **420**. Further, as illustrated in FIG. 12 and FIG. 13, the wiring **480E** is arranged in the center of the Y-axis direction in the first inner layer of the soft substrate **420** so that both ends of the Y-axis positive direction and the Y-axis negative direction are inside the both ends of the soft substrate **420**.

[0141] The wiring **480F** is provided on the outer layer of the soft substrate **420** in the negative X-axis direction, that is, on the surface of the insulating layer **421D** between the insulating layers **421D** and **421E** in the negative Z-axis direction. For example, as illustrated in FIG. 11, the wiring **480F** is provided in the outer layer of the soft substrate **420** in the negative X-axis direction in a range from the positive X-axis end to the negative X-axis end of the soft substrate **420**. As illustrated in FIG. 12 and FIG. 13, the wiring **480F** is arranged in the middle of the Y-axis direction in the outer layer of the soft substrate **420** in the negative X-axis direction, so that both ends of the positive Y-axis direction and the negative Y-axis direction are inside the both ends of the soft substrate **420**. The wiring **480F** may be arranged so that the positions of both ends in the positive Y-axis direction and the negative Y-axis direction approximately coincide with the wiring **480E**.

[0142] The electrical connection between the wirings **480E** and **480F** and the pair of hard substrates **410a** is implemented by a connection part **483a**. For example, as illustrated in FIG. 11 and FIG. 13, the connection parts **483a** are provided at the center in the Y-axis direction and toward the positive X-axis direction than the center in the X-axis direction of the pair of hard substrates **410a**, and each connects the wirings **480E** and **480F** and the pair of hard substrates **410a** in the Z-axis direction.

[0143] The electrical connection between wirings **480E** and **480F** and the pair of hard substrates **410b** is implemented by a connection part **483b**. For example, as illustrated in FIG. 11 and FIG. 13, two connection parts **483b** are provided in the X-axis direction at the center of the Y-axis direction of the pair of hard substrates **410b**, and each connects the wirings **480E** and **480F** and the pair of hard substrates **410b** in the Z-axis direction.

[0144] The electrical connection between wirings **480E** and **480F** and the pair of hard substrates **410c** is implemented by a connection part **483c**. For example, as illustrated in FIG. 11 and FIG. 13, the connection parts **483c** are provided at the center in the Y-axis direction and toward the negative X-axis direction than the center in the X-axis direction of the pair of hard substrates **410c**, and each

connects the wirings **480E** and **480F** and the pair of hard substrates **410c** in the Z-axis direction.

[0145] The connection part **483a** may achieve electrical connection between the wirings **480E** and **480F** and the pair of hard substrates **410a** by any method. The connection part **483a** implements the electrical connection between the wirings **480E** and **480F** and the pair of hard substrates **410a** by, for example, via connection. Further, the connection part **483a** may implement an electrical connection between the wirings **480E** and **480F** and the pair of hard substrates **410a** by copper inlay or the like. The same may be applied to the connection parts **483b** and **483c**.

[0146] In this example, the wiring **490** includes wirings **490I** and **490J** as a portion provided on the soft substrate **420**.

[0147] The wiring **490I** is provided on the outer layer of the soft substrate **420** in the positive X-axis direction, that is, on the surface of the insulating layer **421B** in the positive Z-axis direction between the insulating layers **421A** and **421B**. For example, as illustrated in FIG. **11**, the wiring **490I** is provided in the outer layer of the soft substrate **420** in the positive X-axis direction in a range from the positive X-axis direction end to the negative X-axis direction end of the soft substrate **420**. For example, as illustrated in FIG. **12** and FIG. **13**, the wiring **490I** is arranged in the middle of the Y-axis direction in the outer layer of the soft substrate **420** in the positive X-axis direction so that both ends of the positive Y-axis direction and the negative Y-axis direction are inside the both ends of the soft substrate **420**. The wiring **490I** may be arranged so that the positions of both ends in the positive Y-axis direction and the negative Y-axis direction approximately coincide with the wirings **480E** and **480F**.

[0148] The wiring **490J** is provided on the second inner layer of the soft substrate **420**, that is, on the surface of the insulating layer **421C** in the negative Z-axis direction between the insulating layers **421C** and **421D**. For example, as illustrated in FIG. **11**, in the second inner layer of the soft substrate **420**, the wiring **490J** is provided in the range from the positive X-axis end to the negative X-axis end of the soft substrate **420**. For example, as illustrated in FIG. **12** and FIG. **13**, in the second inner layer of the soft substrate **420**, the wiring **490J** is arranged in the center of the Y-axis direction so that both ends of the positive and negative Y-axis directions are inside the both ends of the soft substrate **420**. The wiring **490I** may be arranged so that the positions of both ends in the positive Y-axis direction and the negative Y-axis direction approximately coincide with the wirings **480E** and **480F**.

[0149] The electrical connection between the wirings **490I** and **490J** and the pair of hard substrates **410a** is implemented by a connection part **493a**. For example, as illustrated in FIG. **11** and FIG. **13**, the connection parts **493a** are provided at both ends of the wirings **490I** and **490J** in the Y-axis direction and toward the negative X-axis direction than the center of the pair of hard substrates **410a** in the X-axis direction, and each connects the wirings **490I** to **490J** and the pair of hard substrates **410a** in the Z-axis direction.

[0150] The electrical connection between the wirings **490I** and **490J** and the pair of hard substrates **410b** is implemented by a connection part **493b**. For example, as illustrated in FIG. **11**, the connection parts **493b** are provided at both ends of the pair of hard substrates **410b** in the X-axis direction, and each connects the wirings **490I** and **490J** and the pair of hard substrates **410b** in the Z-axis direction. For example, as illustrated in FIG. **13**, the connection parts **493b** are provided at both ends of the wirings **490I** and **490J** in the Y-axis direction, and each connects the wirings **490I** and **490J** and the pair of hard substrates **410b** in the Z-axis direction.

[0151] The electrical connection between the wirings **490I** to **490J** and the pair of hard substrates **410c** is implemented by the connection part **493c**. For example, as illustrated in FIG. **11** and FIG. **13**, the connection part **493c** is provided at both ends of the wirings **490I** and **490J** in the Y-axis direction and toward the positive X-axis direction than the center of the pair of hard substrates **410c** in the X-axis direction, and connects the wirings **490I** to **490J** and the pair of hard substrates **410c** in the Z-axis direction.

[0152] The connection part **493a** may achieve electrical connection between the wirings **490I** and

490J and the pair of hard substrates **410a** by any method. The connection part **493a** implements electrical connection between the wirings **490I** and **490D** and the pair of hard substrates **410a** by, for example, via connection. Further, the connection part **493a** may implement an electrical connection between the wirings **490I** and **490J** and the pair of hard substrates **410a** by copper inlay or the like. The same may be applied to the connection parts **493b** and **493c**.

[0153] In this way, the wirings **480** and **490** are alternately arranged in the Z-axis direction, that is, in the stacking direction of the wiring layers, in the soft substrate **420**. Thus, a part of the parasitic inductance component of the wirings **480** and **490** can be canceled. Therefore, the inductance per unit wiring length of the wirings **480** and **490** can be reduced. Therefore, the gate oscillation phenomenon caused by the parasitic component of the wirings **480** and **490** and the imbalance of the current flowing through the semiconductor switch element SW can be prevented, and the wiring board **400** with higher reliability and operation stability can be implemented.

Sixth Example of Wiring Board

[0154] Next, a sixth example of the wiring board **400** according to the present embodiment will be described with reference to FIG. **14**.

[0155] The following discussion will focus on the parts that differ from the first to fifth examples above, and the same or corresponding content as the first to fifth examples above may be omitted.

[0156] FIG. **14** is a circuit diagram illustrating an example of the power conversion apparatus **1** including a sixth example of the wiring board **400**.

[0157] In FIG. **14**, unlike FIGS. **1** and **4**, the main circuit is omitted and only the arm part **10** is drawn. The same applies to FIGS. **15** to **18** below.

[0158] As illustrated in FIG. **14**, this example differs from the above second example in that common-mode choke coils **495a** to **495c** are added, and is otherwise the same as the above second example.

[0159] The wiring board **400** includes the common-mode choke coils **495a** to **495c**.

[0160] The common-mode choke coil **495a** is provided to straddle the wiring **480a** connecting between the electrodes **450a** and **460a** and the wiring **490a** connecting between the electrodes **450b** and **470a**. Specifically, the coil on the primary-side of the common-mode choke coil **495a** is arranged in the form of being inserted into the wiring **480a**, and the coil on the secondary-side of the common-mode choke coil **495a** is arranged in the form of being inserted into the wiring **490a**. For example, as illustrated in FIG. **14**, the common-mode choke coil **495a** is arranged on the hard substrate **430a**.

[0161] The common-mode choke coil **495b** is provided to straddle the wiring **480b** connecting between the electrodes **450a** and **460b** and the wiring **490b** connecting between the electrodes **450b** and **470b**. Specifically, the coil on the primary-side of the common-mode choke coil **495b** is arranged on the wiring **480b**, and the coil on the secondary-side of the common-mode choke coil **495b** is arranged on the wiring **490b**. For example, as illustrated in FIG. **14**, the common-mode choke coil **495b** is arranged on the hard substrate **430b**.

[0162] The common-mode choke coil **495c** is provided to straddle the wiring **480c** connecting between the electrodes **450a** and **460c** and the wiring **490c** connecting between the electrodes **450b** and **470c**. More specifically, a coil on the primary-side of the common-mode choke coil **495c** is arranged on the wiring **480c**, and a coil on the secondary-side of the common-mode choke coil **495c** is arranged on the wiring **490c**. For example, as illustrated in FIG. **14**, the common-mode choke coil **495c** is arranged on a hard substrate **430c**.

[0163] Thus, in this example, the common-mode choke coils **495a** to **495c** are provided so as to straddle the wirings **480** the **490**. Thus, by reducing the common-mode current between the semiconductor modules **100a** to **100c**, the cross current (circulating current) of the wirings **480** and **490** caused by the variation in the characteristics of the semiconductor switch element SW for each of the semiconductor modules **100a** to **100c** can be reduced. Therefore, it is possible to reduce the gate oscillation phenomenon and the current imbalance and the like for each of the semiconductor

modules **100a** to **100c**, while mitigating the degradation of the switching performance of the semiconductor switch element SW. Therefore, the wiring board **400** with higher reliability and operation stability can be implemented.

Seventh Example of Wiring Board

[0164] Next, a seventh example of the wiring board **400** according to the present embodiment will be described with reference to FIG. **15**.

[0165] The following discussion will focus on the parts that differ from first to sixth examples, and the same or corresponding content as the first to sixth examples above may be omitted.

[0166] FIG. **15** is a circuit diagram illustrating an example of the power conversion apparatus **1** including a seventh example of the wiring board **400**.

[0167] As illustrated in FIG. **15**, this example differs from the above second example in that magnetic cores **496a** to **496c** are added, and is otherwise the same as the above second example.

[0168] The wiring board **400** includes the magnetic cores **496a** to **496c** (an example of a magnetic core for common-mode choke).

[0169] For example, as illustrated in FIG. **15**, the magnetic core **496a** has an approximately annular shape and is provided to enclose the wiring **480a** connecting between the electrodes **450a** and **460a** and the wiring **490a** connecting between the electrodes **450b** and **470a**. For example, as illustrated in FIG. **15**, the magnetic core **496a** is arranged on the hard substrate **430a**.

[0170] For example, as illustrated in FIG. **15**, the magnetic core **496b** has an approximately annular shape and is provided to enclose the wiring **480b** connecting between the electrodes **450a** and **460b** and the wiring **490b** connecting between electrodes **450b** and **470b**. For example, as illustrated in FIG. **15**, the magnetic core **496b** is arranged on the hard substrate **430b**.

[0171] For example, as illustrated in FIG. **15**, the magnetic core **496c** has an approximately annular shape and is provided to surround the wiring **480c** connecting between the electrodes **450a** and **460c** and the wiring **490c** connecting between the electrodes **450b** and **470c**. For example, as illustrated in FIG. **15**, the magnetic core **496c** is arranged on the hard substrate **430c**.

[0172] The magnetic cores **496a** to **496c** may be configured as one unit or in a divisible form from the viewpoint of ensuring mechanical strength and mountability on the wiring board **400**.

[0173] Thus, in this example, the magnetic cores **496a** to **496c** are provided to surround the wirings **480** and **490**. Thus, similar to the common-mode choke coils **495a** to **495c** of the sixth example above, the common mode current can be prevented, and as a result, the cross current (circulating current) of the wirings **480** and **490** caused by the variation in the characteristics of the semiconductor switch element SW for each of the semiconductor modules **100a** to **100c** can be reduced. Therefore, it is possible to prevent the gate oscillation phenomenon and the current imbalance and the like for each of the semiconductor modules **100a** to **100c**, while mitigating the degradation of the switching performance of the semiconductor switch element SW. Therefore, the wiring board **400** with higher reliability and operation stability can be implemented.

Eighth Example of Wiring Board

[0174] Next, an eighth example of the wiring board **400** according to the present embodiment will be described with reference to FIG. **16**.

[0175] The following discussion will focus on the parts that differ from the first to seventh examples above, and the same or corresponding content as the first to seventh examples above may be omitted.

[0176] FIG. **16** is a circuit diagram illustrating an example of the power conversion apparatus **1** including an eighth example of the wiring board **400**.

[0177] As illustrated in FIG. **16**, this example differs from the above second example in that magnetic cores **497a** and **497b** are added, and is otherwise the same as the above second example.

[0178] The wiring board **400** includes the magnetic cores **497a** and **497b** (an example of a magnetic core for common-mode choke).

[0179] For example, as illustrated in FIG. **16**, the magnetic core **497a** has an approximately annular

shape and is provided to surround the soft substrate **440a** on which the wiring **480** (the wiring **480a**) and the wiring **490** (the wiring **490a**) are arranged. The magnetic core **497a** is held and fixed to at least one of, for example, the hard substrate **430a** and the hard substrate **430b**. Any configuration may be adopted for the holding mechanism (fixing mechanism) between the magnetic core **497a** and the hard substrates **430a** and **430b**. For example, a cutout is provided in the magnetic core **497a** and the magnetic core **497a** is held and fixed in such a way that and the hard substrates **430a** and **430b** are inserted in the cutout.

[0180] Part of the wirings **480b** and **490b** may be arranged on the soft substrate **440a**.

[0181] For example, as illustrated in FIG. **16**, the magnetic core **497b** has an approximately annular shape and is provided to surround the soft substrate **440a** on which the wiring **480** (the wiring **480c**) and the wiring **490** (the wiring **490c**) are arranged. The magnetic core **497b** is held and fixed to at least one of, for example, the hard substrate **430b** and the hard substrate **430c**. Any configuration may be adopted for the holding mechanism (fixing mechanism) between the magnetic core **497a** and the hard substrates **430b** and **430c**. For example, a cutout is provided in the magnetic core **497b** and the magnetic core **497b** is held and fixed in such a way that and the hard substrates **430b** and **430c** are inserted in the cutout.

[0182] Part of the wirings **480b** and **490b** may be arranged on the soft substrate **440b**.

[0183] The magnetic cores **497a** and **497b** may be configured as one body or in a dividable form, similar to the magnetic cores **496a** to **496c** in the seventh example above.

[0184] Thus, in this example, the magnetic cores **497a** and **497b** are provided to surround the soft substrates **440a** and **440b** on which the wirings **480** and **490** are provided. Thus, as in the cases of the sixth and seventh examples above, the common-mode current can be prevented, and as a result, the cross current (circulating current) in the wirings **480** and **490** caused by the variation in the characteristics of the semiconductor switch element SW can be reduced for each of the semiconductor modules **100a** to **100c**. Therefore, it is possible to prevent the gate oscillation phenomenon and the current imbalance and the like for each of the semiconductor modules **100a** to **100c**, while mitigating the degradation of the switching performance of the semiconductor switch element SW. Therefore, the wiring board **400** with higher reliability and operation stability can be implemented.

Ninth Example of Wiring Board

[0185] Next, a ninth example of the wiring board **400** according to the present embodiment will be described with reference to FIG. **17**.

[0186] The following discussion will focus on the parts that differ from the first to eighth examples above, and the same or corresponding content as the first to eighth examples above may be omitted.

[0187] FIG. **17** is a circuit diagram illustrating an example of the power conversion apparatus **1** including a ninth example of the wiring board **400**.

[0188] As illustrated in FIG. **17**, this example differs from the second example above in that resistors **498a** to **498c** are added, and is otherwise the same as the second example above.

[0189] The wiring board **400** includes the resistors **498a** to **498c** (an example of the first resistor).

[0190] The resistor **498a** is arranged in such a way as to be inserted into the wiring **480a** connecting between the electrodes **450a** and **460a**. For example, as illustrated in FIG. **17**, the resistor **498a** is arranged on the hard substrate **430a**.

[0191] The resistor **498b** is arranged in such a way as to be inserted into the wiring **480b** connecting between the electrodes **450a** and **460b**. For example, as illustrated in FIG. **17**, the resistor **498b** is arranged on the hard substrate **430b**.

[0192] The resistor **498c** is arranged in such a way as to be inserted into the wiring **480c** connecting between the electrodes **450a** and **460c**. For example, as illustrated in FIG. **17**, the resistor **498c** is arranged on the hard substrate **430c**.

[0193] The resistance values of the resistors **498a** to **498c** are set appropriately and may be approximately the same as each other, or at least some of these values may be different.

[0194] Thus, in this example, the resistors **498a** to **498c** are arranged on the wiring **480**. This can prevent a current imbalance caused by variations and the like in the characteristics of the semiconductor switch elements SW for each of the semiconductor modules **100a** to **100c**. When the current imbalance is negligible, the cross current (circulating current) between the semiconductor modules **100a** to **100c** can be reduced. This is particularly suitable when it is not possible to secure space for mounting the magnetic cores **496a** to **496c** of the seventh example and the magnetic cores **497a** and **497b** of the eighth example.

Tenth Example of Wiring Board

[0195] Next, a tenth example of the wiring board **400** according to the present embodiment will be described with reference to FIG. **18**.

[0196] The following discussion will focus on the parts that differ from the first to ninth examples above, and the same or corresponding content as the first to ninth examples above may be omitted.

[0197] FIG. **18** is a circuit diagram illustrating one example of the power conversion apparatus **1** including the tenth example of the wiring board **400**.

[0198] As illustrated in FIG. **18**, this example differs from the second example above in that resistors **499a** to **499c** are added, and is otherwise the same as the second example above.

[0199] The wiring board **400** includes the resistors **499a** to **499c** (an example of a second resistor).

[0200] The resistor **499a** is arranged in such a way as to be inserted into the wiring **490a** connecting between the electrodes **450b** and **470a**. For example, as illustrated in FIG. **18**, the resistor **499a** is arranged on the hard substrate **430a**.

[0201] The resistor **499b** is arranged in such a way as to be inserted into the wiring **490b** connecting between the electrodes **450b** and **470b**. For example, as illustrated in FIG. **18**, the resistor **499b** is arranged on the hard substrate **430b**.

[0202] The resistor **499c** is arranged in such a way as to be inserted into the wiring **490c** connecting between the electrodes **450b** and **470c**. For example, as illustrated in FIG. **18**, the resistor **499c** is arranged on the hard substrate **430c**.

[0203] The resistance values of the resistors **499a** to **499c** are set appropriately and may be approximately the same as each other, or at least some of these values may be different.

[0204] Thus, in this example, the resistors **499a** to **499c** are arranged on the wiring **490**. This can prevent the current imbalance caused by variations and the like in the characteristics of the semiconductor switch elements SW for each of the semiconductor modules **100a** to **100c**. When the current imbalance is negligible, the cross current (circulating current) between the semiconductor modules **100a** to **100c** can be prevented. This is particularly suitable when it is not possible to secure space for mounting the magnetic cores **496a** to **496c** of the seventh example and the magnetic cores **497a** and **497b** of the eighth example.

Eleventh Example of Wiring Board

[0205] Next, an eleventh example of the wiring board according to the present embodiment will be described.

[0206] In this example, the circuit diagram of the power conversion apparatus **1** including the wiring board **400** is the same as in the first example described above, so the illustration is omitted and the explanation is given by referring to FIG. **1**.

[0207] This example differs from the above first example in that the structures of the electrode **450** and the electric circuit **500** are specified, and is otherwise the same as the above example 1.

[0208] In this example, a coaxial connector is used to connect the wiring **480** and the wiring **490** to the electric circuit **500**. The coaxial connector is, for example, a BNC connector. For example, the electrode **450** connected to the wirings **480** and **490** is a jack (female) of the coaxial connector, and the tip connected to the electrode **450** of the electrical circuit **500** is a plug (male) of the coaxial connector. With this, the plug of the coaxial connector at the tip of the electric circuit **500** can be inserted into the jack of the coaxial connector corresponding to the electrode **450**, and the wiring **480** and the wiring **490** can be connected to the electric circuit **500**.

[0209] Specifically, the wiring **480** is connected to the electric circuit main body (internal conductor) at the center of the coaxial connector (jack), and the wiring **490** is connected to the external conductor surrounding the electric circuit main body of the coaxial connector (jack). That is, the electric circuit main body (internal conductor) at the center of the coaxial connector (jack) as the electrode **450** corresponds to the electrode **450a**, and the external conductor of the coaxial connector (jack) as the electrode **450** corresponds to the electrode **450b**.

[0210] The electric circuit **500** has a coaxial structure including an electric circuit main body at the center (inner conductor) and an outer conductor surrounding the electric circuit main body. That is, the electric circuit main body at the center corresponds to the electric circuit **500a**, and the external conductor surrounding the electric circuit main body corresponds to the electric circuit **500b**. The electric circuit **500** is, for example, a coaxial cable.

[0211] Thus, in this example, a coaxial structure is adopted for the electrode **450** and the electric circuit **500**. In this example, the electric circuit **500** and the wirings **480** and **490** may be connected to each other via a coaxial connector. Specifically, the electric circuit **500a**, which is the electric circuit main body of the electric circuit **500**, and the wiring **480** connected to the electric circuit main body of the coaxial connector including the electrode **450a** are connected; and the electric circuit **500b**, which is the external conductor of the electric circuit **500**, and the wiring **490** connected to the external conductor of the coaxial connector including the electrode **450b** are connected. As a result, the parasitic inductance component of the electrode **450** and the electric circuit **500** in the wiring path from the driving circuit of the driving circuit board **300** to the respective semiconductor switch elements SW of the semiconductor modules **100a** to **100c** can be reduced. Therefore, as in the cases of the third and fourth examples above, the wiring board **400** with higher reliability and operation stability can be implemented.

Twelfth Example of Wiring Board

[0212] Next, a twelfth example of the wiring board **400** will be described with reference to FIGS. **19** and **20**.

[0213] The following discussion will focus on the parts that are different from those of the first to tenth examples above, and the same or corresponding descriptions as those of the first to tenth examples above may be omitted.

[0214] FIG. **19** is a top view illustrating a wiring pattern **480WP** of the eleventh example of the wiring board **400**. Specifically, the figure illustrates an example of the wiring pattern **480WP** of the wiring **480**. FIG. **20** is a top view illustrating a wiring pattern **480WP_C** according to the comparative example.

[0215] In this example, the circuit diagram of the power conversion apparatus **1** including the wiring board **400** is the same as that of the first example described above, so the illustration is omitted and the explanation is given by referring to FIG. **1**.

[0216] The wiring pattern **480WP** electrically connects the electrode **450a** to each of the electrodes **460a** to **460c**. The wiring pattern **480WP** is provided on the soft substrate **420**, for example. The wiring pattern **480WP** may be provided to span across the soft substrate **420** and the hard substrates **410a** to **410c**.

[0217] As illustrated in FIG. **19**, the electrodes **460a** to **460c** are arranged so that their positions in the Y-axis direction are approximately the same with each other and the distance L_a between the electrodes **460a** and **460b** and the distance L_c between the electrodes **460b** and **460c** are approximately the same.

[0218] The electrode **450a** is arranged so that the position thereof in the X-axis direction approximately coincides with that of the electrode **460b**.

[0219] Slits **484** are provided in the wiring pattern **480WP**. Accordingly, a portion of the slit **484** is electrically cut off. Therefore, the wiring length between the electrode **450a** and each of the electrodes **460a** to **460c** can be adjusted to bypass the slit **484**.

[0220] In this example, the slits **484** are provided so that the paths between electrodes **450a** and

460b are two paths that pass through the electrodes **460a** and **460c**, respectively. In this example, the slits **484** are provided so that a wiring length WLa1 between the electrodes **450a** and **460a**, a wiring length WLa1 between the electrodes **450a** and **460c**, a wiring length WLab1 between the electrodes **460a** and **460b**, and a wiring length WLbc1 between the electrodes **460b** and **460c** are approximately the same.

[0221] Therefore, the relationship of the following equation (1) is established between a parasitic inductance component Lpa in the path between the electrodes **450a** and **460a** and the parasitic inductance component Lpc in the path between the electrodes **450a** and **460c**.

$$L_{pa} \approx L_{pc} \quad (1)$$

[0222] Further, the two paths between the electrodes **450a** and **460b**, i.e., the path via the electrode **460a** and the path via the electrode **460c**, have approximately the same length, and the wiring length thereof is twice as long as the path between the electrodes **450a** and **460a**. Therefore, the parasitic inductance component Lpb in the path between the electrodes **450a** and **460b** satisfy the relationship of the following equation (2).

$$L_{pb} \approx 2 \cdot L_{pa} / 2 \cdot L_{pa} = L_{pa} \quad (2)$$

[0223] Therefore, as can be seen from equations (1) and (2), in this example, by providing the slit **484**, the inductance components of the paths between the electrode **450a** and each of the electrodes **460a** to **460c** can be approximately equalized.

[0224] For example, when the semiconductor module **100b** is omitted and the wiring board **400** drives two of the semiconductor modules **100a** and **100c**, the wiring lengths WLa1 and WLa1 can be matched to be approximately the same relatively easily as illustrated in FIG. 20. However, when there are three or more semiconductor modules driven by the wiring board **400**, and an attempt is made to match the wiring lengths only by the geometric arrangement of the electrodes **460a** to **460c**, this may result in an increase in the mounting area of the wiring pattern **480WP** or an increase in unnecessary wiring lengths.

[0225] In contrast, in the present example, the slit **484** is provided in the wiring pattern **480WP** of the wiring **480** so that the inductance components of the paths between the electrode **450a** and each of the electrodes **460a** to **460c** are approximately the same. Thus, an increase in parasitic inductance components due to the wiring **480** can be prevented while an inadvertent increase in the mounting area and wiring length of the wiring **480** can be prevented.

Other Examples of Wiring Board

[0226] The above first to eleventh examples of the wiring board may be combined as appropriate.

[0227] For example, the configuration of the wirings **480** and **490** on the hard substrates **430a** to **430c** of the third example described above may be applied to the hard substrates **410a** to **410c** of the first, eleventh, and twelfth examples described above.

[0228] Further, for example, the arrangement structure of the wirings **480** and **490** in the soft substrate **420** of the fourth and fifth examples described above may be applied to the soft substrates **440a** and **440b** of the second, third, and sixth to tenth examples described above.

[0229] Further, for example, the arrangement structure of the wirings **480** and **490** in the soft substrate **420** of the fifth example described above may be applied to the hard substrates **410a** to **410c** or the hard substrates **430a** to **430c** of the first, second, fourth, sixth to twelfth examples and the variations thereof described above.

[0230] Further, for example, the common-mode choke coils **495a** to **495c** of the sixth example described above may be applied to the wirings **480** and **490** of the first, eleventh, and twelfth examples and the variations thereof described above. The same may be applied to the magnetic cores **496a** to **496c** of the seventh example described above, the resistors **498a** to **498c** of the ninth example described above, and the resistors **499a** to **499c** of the tenth example described above.

[0231] Further, for example, the magnetic cores **497a** and **497b** of the eighth example described

above may be applied to the soft substrates **420** of the first, eleventh, and twelfth examples and the variations thereof described above.

[0232] Further, for example, the aforementioned the ninth example or the variations thereof and the aforementioned tenth example or the variations thereof may be combined. That is, both resistors **498a** to **498c** and resistors **499a** to **499c** may be provided on the wiring board **400**.

[0233] Further, for example, the structures of the electrode **450** and the electric circuit **500** in the eleventh example described above may be applied to the second to tenth and twelfth examples and the variations thereof described above.

[0234] For example, the slit **484** of the wiring pattern **480WP** of the twelfth example may be applied to the wiring pattern of the wiring **480** of the second to eleventh examples and the variations thereof.

[0235] The first to eleventh examples of the above wiring boards and the variations thereof may be modified or changed as appropriate.

[0236] For example, in the first to eleventh examples and the variations thereof above, the number of semiconductor switch elements SW (corresponding semiconductor modules) connected in parallel that are the driving targets of the driving circuit board **300** and the wiring board **400**, may be two or four or more. In this case, by adopting the same configuration and structure as those of the first to eleventh examples described above and the variations thereof, the same function and effect can be achieved.

[0237] For example, in the above twelfth example and the variations thereof, the number of semiconductor switch elements SW (corresponding semiconductor modules) connected in parallel that are the driving targets of the driving circuit board **300** and the wiring board **400** may be four or more. In this case, by adopting the same configuration and structure (the slit **484**) as in the twelfth example described above and the variations thereof, the same function and effect can be achieved.

[0238] Further, for example, in the above third example and the variations thereof, the number of wiring layers to be stacked on the hard substrates **410a** to **410c** or the hard substrates **430a** to **430c** may be three or five or more. In this case, by adopting the same configuration and structure (the stacked structure of the wirings **480** and **490**) as in the above third example and the variations thereof, the same operation and effect are achieved.

[0239] Further, for example, in the above third example and the variations thereof, instead of the connection part **491**, other electrical connection methods related to the wiring **490** of each wiring layer of the hard substrates **430a** to **430c** and the soft substrates **440a** and **440b** may be used, in order to electromagnetically shield the wiring **480**. For example, the same effect can be obtained by adopting an electrical connection method such as plating the ends of the soft substrates **440a** and **440b** and the hard substrates **430a** to **430c** and connecting to the wiring **490** of each wiring layer and the like.

[0240] Further, for example, in the above fourth example and the variations thereof, instead of the connection parts **492a** to **492c**, other electrical connection methods related to the wiring **490** of each wiring layer of the hard substrate **410a** to **410c** or the soft substrate **420** may be used, in order to electromagnetically shield the wiring **480**. For example, by adopting an electrical connection method such as plating the ends of the soft substrate **420** and the hard substrates **410a** to **410c** and connecting to the wiring **490** of each wiring layer and the like, the same effect can be obtained.

[0241] Further, for example, in the fourth and fifth examples and the variations thereof above, the number of wiring layers to be stacked in the soft substrate **420** or the soft substrates **440a** and **440b** may be three or five or more. In this case, by adopting the same configuration and structure (a stacked structure of the wirings **480** and **490**) as in the fourth example above and the variations thereof, the same function and effect are achieved.

[0242] Further, for example, in the above eleventh example and the variations thereof, instead of a coaxial structure, the electric circuit **500** may have a structure of a soft substrate and wiring corresponding to the electric circuits **500a** and **500b** on the soft substrate. In this case, the wiring of

the soft substrate corresponding to the electric circuits **500a** and **500b** may have the same arrangement structure as the wirings **480** and **490** in the soft substrate **420** of the above third example. Accordingly, the same function and effect as the case of a coaxial structure are achieved. [0243] Further, for example, in the first to twelfth examples and variations thereof, the wiring board **400** may be configured to drive multiple sets of multiple semiconductor modules (the semiconductor switch elements SW) connected in parallel. For example, this is suitable when a 2-in-1 package semiconductor module including two semiconductor switch elements SW corresponding to the upper and lower arms is adopted. In this case, multiple sets of the electrodes **460a** to **460c**, the electrodes **470a** to **470c**, and the wirings **480** and **490** for driving a specific single set of multiple semiconductor modules are mounted on the hard or soft substrate of the wiring board **400**.

Functions

[0244] Next, the functions of the wiring board **400** according to the present embodiment will be described.

[0245] In the present embodiment, the wiring board **400** includes a pair of hard substrates (for example, the hard substrates **410a** to **410c**), a soft substrate (for example, the soft substrate **420**), a first electrode (for example, the electrodes **460a** to **460c**), a second electrode (for example, the electrodes **470a** to **470c**), a first wiring (for example, the wiring **480**), and a second wiring (for example, the wiring **490**). Specifically, a pair of hard substrates is provided for each of a plurality of semiconductor elements (e.g., semiconductor switch elements SW) connected in parallel. Further, the soft substrate is provided so that at least part thereof is sandwiched between all the pairs of hard substrates. The first electrode connects the control terminal of the semiconductor element to the hard or soft substrate. The second electrode connects the reference potential terminal of the semiconductor element to the hard or soft substrate. Further, at least a part of the first wiring is provided on the soft substrate, and the first electrodes for each of the multiple semiconductor elements are connected in parallel. The second wiring, at least partially provided on the soft substrate, connects in parallel the second electrodes for each of the multiple semiconductor elements.

[0246] Further, in the present embodiment, the wiring board **400** includes a hard substrate (for example, the hard substrates **430a** to **430c**), a soft substrate (for example, the soft substrates **440a** and **440b**), a first electrode (for example, the electrodes **460a** to **460c**), a second electrode (for example, the electrodes **470a** to **470c**), a first wiring (for example, the wiring **480**), and a second wiring (for example, the wiring **490**). Specifically, the hard substrate is provided for each of the multiple semiconductor elements (for example, the semiconductor switch element SW) connected in parallel. Further, the soft substrate connects between the hard substrates. The first electrode connects the control terminal of the semiconductor element to the hard substrate. The second electrode connects the reference potential terminal of the semiconductor element to the hard substrate. The first wiring is provided on the hard and soft substrates, and connects the first electrodes for each of the multiple semiconductor elements in parallel. The second wiring is provided on the hard and soft substrates, and connects the second electrodes for each of the multiple semiconductor elements in parallel. Further, the hard substrates for each of the multiple semiconductor elements are all connected by the soft substrate.

[0247] Thus, the wiring length between the driving circuit and the control terminal for each semiconductor element can be adjusted by the wiring pattern in the hard and soft substrates. Therefore, the degree of freedom of the wiring pattern can prevent an increase in the wiring length when matching the wiring length between the driving circuit and the control terminal for each semiconductor element. As a result, the performance of the semiconductor element can be brought to a desired level and the occurrence of the gate oscillation phenomenon can be reduced. Further, connectors are not used for the interconnection of the hard substrates, and, therefore, the occurrence of connection failure problems can be reduced and the height of the wiring board **400**

can be reduced. Therefore, the wiring board **400** with higher reliability and operation stability can be implemented. Moreover, a manufacturing process for fitting the connector is not required, and labor saving in assembly work, improvement in manufacturing efficiency, and reduction in manufacturing cost can be achieved.

[0248] Further, in the present embodiment, the hard substrate may have three or more wiring layers. The first wiring may be provided in the inner layer among the three or more wiring layers in the hard substrate. The second wiring may be provided on both of the outer layers among the three or more wiring layers in the hard substrate, and may also be provided on the inner layer so as to sandwich the first wiring therebetween.

[0249] This allows the first wiring to be electromagnetically shielded by the second wiring, thereby preventing (canceling) the parasitic inductance component of the second wiring of the hard substrate. Therefore, it is possible to prevent the gate oscillation phenomenon caused by the parasitic components of the first and second wirings and the imbalance of the current flowing to the semiconductor element, and to implement the wiring board **400** with higher reliability and operation stability.

[0250] Further, in the present embodiment, in the wiring board **400**, the soft substrate may have three or more wiring layers. The first wiring may be provided in the inner layer among three or more wiring layers in the soft substrate. The second wiring may be provided on both of the outer layers among the three or more wiring layers in the soft substrate, and may also be provided on the inner layer so as to sandwich the first wiring therebetween.

[0251] This allows the first wiring to be electromagnetically shielded by the second wiring, thereby preventing (canceling) the parasitic inductance component of the second wiring of the soft substrate. Therefore, it is possible to prevent the gate oscillation phenomenon caused by the parasitic components of the first and second wirings and the imbalance of the current flowing through the semiconductor switch element SW to implement the wiring board **400** with higher reliability and operation stability.

[0252] Further, in the present embodiment, the first wiring and the second wiring may be alternately stacked on at least one of the hard substrate or the soft substrate.

[0253] Thus, part of the parasitic inductance components of the first and second wirings can be canceled. Therefore, the inductance per unit wiring length of the first and second wirings can be reduced. Therefore, it is possible to prevent the gate oscillation phenomenon caused by the parasitic components of the first and second wirings and the imbalance of the current flowing through the semiconductor switch element SW to implement the wiring board **400** with higher reliability and operation stability.

[0254] Further, in the present embodiment, the wiring board **400** may be provided with a common-mode choke coil (for example, the common-mode choke coils **495a** to **495c**) including a primary-side coil provided in the first wiring and a secondary-side coil provided in the second wiring.

[0255] Thus, by preventing the common-mode current between semiconductor elements, the cross current (circulating current) in the first and second wirings, which is caused by variations in the characteristics of each semiconductor element, can be reduced. Therefore, it is possible to prevent the gate oscillation phenomenon, the current imbalance of each semiconductor element, etc., while mitigating the degradation of the switching performance of the semiconductor element. Therefore, the wiring board **400** with higher reliability and operation stability can be implemented.

[0256] Further, in the present embodiment, the wiring board **400** may be provided with a magnetic core (for example, the magnetic cores **496a** to **496c** and the magnetic cores **497a**, **497b**) for the common mode choke surrounding the first and second wirings.

[0257] Accordingly, as in the case of the common-mode choke coil, the common-mode current can be prevented, and as a result, the cross current (circulation current) of the first and second wirings, which is caused by variations in the characteristics of each semiconductor element, can be reduced. Therefore, it is possible to prevent the gate oscillation phenomenon, the current imbalance of each

semiconductor element, etc., while mitigating the degradation of the switching performance of the semiconductor element. Therefore, the wiring board **400** with higher reliability and operation stability can be implemented.

[0258] Further, in the present embodiment, in the wiring board **400**, the magnetic core (for example, the magnetic cores **497a**, **497b**) have cutouts and may be fixed to the hard substrate by the cutouts.

[0259] Thus, the magnetic core can be efficiently fixed to the wiring board **400** with a simple structure.

[0260] Further, in the present embodiment, the wiring board **400** may have a first resistor (for example, the resistors **498a** to **498c**) provided in the first wiring.

[0261] This can prevent the current imbalance caused by variations and the like in the characteristics of each semiconductor element. When the current imbalance is at a negligible level, the cross current (circulating current) between the semiconductor elements can be prevented.

[0262] In the present embodiment, the wiring board **400** may also have a second resistor (for example, the resistors **499a** to **499c**) provided on the second wiring.

[0263] This can prevent the current imbalance caused by variations and the like in the characteristics of each semiconductor element. When the current imbalance is at a negligible level, the cross current (circulating current) between the semiconductor elements can be prevented.

[0264] Further, in the present embodiment, the wiring board **400** may be provided with a third electrode (for example, the electrode **450a**) connecting the first wiring to the driving circuit for driving the multiple semiconductor elements. Also, the third electrode may have a coaxial structure.

[0265] Thus, the parasitic inductance component of the third electrode in the wiring path from the driving circuit to the multiple semiconductor elements can be reduced. Therefore, the gate oscillation phenomenon caused by the parasitic component and the imbalance of the current flowing to the semiconductor element can be prevented, and the wiring board **400** with higher reliability and operation stability can be implemented.

[0266] Further, in the present embodiment, in the wiring board **400**, the third electrode and the driving circuit may be connected by an electric circuit (for example, the electrical circuit **500a**) having a coaxial structure.

[0267] This reduces the parasitic inductance component of the electrical circuit between the third electrode and the driving circuit in the wiring path from the driving circuit to the multiple semiconductor elements. Therefore, the gate oscillation phenomenon caused by the parasitic component and the imbalance of the current flowing to the semiconductor element can be prevented, and the wiring board **400** with higher reliability and operation stability can be implemented.

[0268] Further, in the present embodiment, the wiring board **400** may be provided with a third electrode (For example, the electrode **450a**) connecting the first wiring to the driving circuit for driving the multiple semiconductor elements. Also, a slit (for example, the slit **484**) may be provided in the wiring pattern (for example, the wiring pattern **480WP**) of the first wiring so that the parasitic impedance is equal between the paths of the parallel connection of the third electrode and the first electrode for each of the multiple semiconductor elements.

[0269] Thus, an increase in parasitic inductance components due to the wiring **480** can be prevented while an inadvertent increase in the mounting area and wiring length of the wiring **480** can be prevented.

[0270] In the present embodiment, in the wiring board **400**, the first electrode, the second electrode, the first wiring, and the second wiring may be mounted for each of a plurality of sets, each set of the plurality of sets including the plurality of semiconductor elements.

[0271] Thus, for example, when a 2 in 1 package semiconductor module including two semiconductor elements of upper and lower arms is utilized, the wiring board **400** can drive multiple sets of multiple semiconductor elements connected in parallel.

[0272] While the embodiment of the present disclosure has been described in detail above, the present disclosure is not limited to the specific embodiment described above, and various modifications and variations can be made within the scope of the subject matter of the present invention as claimed.

Claims

1. A wiring board comprising: a hard substrate provided for each of a plurality of semiconductor elements connected in parallel; and a soft substrate configured to connect the hard substrates to each other, wherein all of the hard substrates of each of the plurality of semiconductor elements are connected by the soft substrate.
2. The wiring board according to claim 1, wherein the hard substrate includes a plurality of layers, and the soft substrate is provided so as to be sandwiched between the plurality of layers.
3. The wiring board according to claim 1, wherein the plurality of semiconductor elements are connected in parallel by the soft substrate.
4. The wiring board according to claim 1, further comprising: a first electrode configured to connect a control terminal of each of the plurality of semiconductor elements to the hard substrate; a second electrode configured to connect a reference potential terminal of each of the plurality of semiconductor elements to the hard substrate; a first wiring configured to connect in parallel the first electrodes of each of the plurality of semiconductor elements, the first wiring being provided on the hard substrate and the soft substrate; and a second wiring configured to connect in parallel the second electrodes of each of the plurality of semiconductor elements, the second wiring being provided on the hard substrate and the soft substrate.
5. The wiring board according to claim 1, wherein the hard substrate includes three or more wiring layers, the first wiring is provided on an inner layer among the three or more wiring layers in the hard substrate, and the second wiring is provided on both outer layers among the three or more wiring layers and also on the inner layer so as to sandwich the first wiring in the hard substrate.
6. The wiring board according to claim 1, wherein the soft substrate includes three or more wiring layers, the first wiring is provided on an inner layer among the three or more wiring layers in the soft substrate, and the second wiring is provided on both outer layers among the three or more wiring layers and also on the inner layer so as to sandwich the first wiring in the soft substrate.
7. The wiring board according to claim 1, wherein in at least one of the hard substrate or the soft substrate, the first wiring and the second wiring are alternately stacked.
8. The wiring board according to claim 1, further comprising: a common-mode choke coil including a primary-side coil provided on the first wiring and a secondary-side coil provided on the secondary wiring.
9. The wiring board according to claim 1, further comprising: a magnetic core for common-mode choke surrounding the first wiring and the second wiring.
10. The wiring board according to claim 9, wherein the magnetic core includes a cutout, and the magnetic core is fixed to the hard substrate by the cutout.
11. The wiring board according to claim 1, further comprising: a first resistor provided on the first wiring.
12. The wiring board according to claim 1, further comprising: a second resistor provided on the second wiring.
13. The wiring board according to claim 1, further comprising: a third electrode configured to connect a driving circuit and the first wiring, the driving circuit being configured to drive the plurality of semiconductor elements, wherein the third electrode has a coaxial structure.
14. The wiring board according to claim 13, wherein the third electrode and the driving circuit are connected by an electric circuit having a coaxial structure.
15. The wiring board according to claim 1, further comprising: a third electrode configured to

connect a driving circuit and the first wiring, the driving circuit being configured to drive the plurality of semiconductor elements, wherein a slit is provided in a wiring pattern of the first wiring such that a parasitic impedance is equal between paths that connect in parallel the third electrode and the first electrode of each of the plurality of semiconductor elements.

16. The wiring board according to claim 1 wherein the first electrode, the second electrode, the first wiring, and the second wiring are mounted for each of a plurality of sets, each set of the plurality of sets including the plurality of semiconductor elements.

17. A power conversion apparatus comprising: the wiring board according to claim 1; and the plurality of semiconductor elements.
